Project Overview:

Author: Haari Muthukumar

As someone interested in solar energy and sustainability, I was curious to see how this impacted our everyday communities and the power that we get. I have created a single-line diagram model and am modeling the power flow in Python to get a grasp of the concepts of grid protection and grid integration with renewable energy and battery storage systems for a self-sustaining microgrid.

SLD:

Voltage Design Choices:

Voltage levels:

This design used a 13.8kV voltage line as the utility line, as this represents the standard distribution voltage coming into a small neighborhood from the larger macrogrid. It then drops down to 4.16kV in the transformer to travel through the residential neighborhood to reduce the severity of arc flash incidents, as well as reduce economic cost, as the switchbox and components inside, like circuit breakers, are bigger, take up more space, and are more expensive when dealing with higher voltages. The 4.16kV travels throughout the busbar and drops to 480V in the branches that have the Solar Panel and the BESS. Many standard BESS and Panel systems around the 1-2 MVA area only have a maximum output voltage of 480V. They also output only DC voltage, and since the grid has AC voltage, we must convert the DC to AC with an inverter. This then gets bumped up to 4.16kV using a transformer, which then goes back to the busbar. Since the BESS also stores power, we have a bidirectional inverter that converts DC to AC and AC to DC as the BESS stores and discharges power. When storing, the voltage drops from 4.16kV to 480V to store in the BESS as well. The final branch, the residential load, drops the 4.16kV from the busbar to around 120-240 V, the standard residential voltage.

Phase Levels:

This grid uses a 3-phase system to provide continuous power to the localized grid, with a 120-degree phase shift, also giving around 73% more power and using less conductive material, meaning better for longer travel. There is a one-phase branch going to the houses, which allows for lower voltage and hence safer, along with cheaper equipment, leading to a more practical, lower cost.

Design Safety Inclusion:

Circuit Breakers:

As Safety is extremely important, this design includes some safety measures. Circuit breakers were put before and after major components to shut down the current flowing in case of component failure or a wider grid failure to safely island or isolate the microgrid.

ANSI Usage:

This design used ANSI symbols to indicate the function of the circuit breakers. Most have standard overcurrent protection, instantaneous and time-delayed (ANSI 50/51), from short circuits or components failing and creating too much current. However, near the voltage sources like the BESS and solar panel, additional protection was required. This design includes overvoltage and undervoltage protection (ANSI 27/59) to make sure that the sources stay in compliance with the IEEE standards so that the other components don't get damaged by changes in voltage. Frequency protection was used as well (ANSI 81), as inverters are highly dependent on frequency, as they usually run on a standard frequency like 60 Hz and aren't manufactured to run on different frequencies, and this frequency should be constant throughout the grid. In the main branch connecting the utility, we have all the ANSI numbers used already, along with an additional synchronization component (ANSI 25) making sure the voltages from the neighborhood side match the voltages from the utility side at the point of common coupling when the grid is already islanded due to an outside macrogrid failure, before rejoining the wider grid. This requires PTs and CTs, which are added. ANSI 67 was also used to identify potential directional differences between the utility side and the microgrid residential side, in case of a utility or residential side blackout, preventing backflow into either grid side.

PT CT Usage:

For this design, I have two PT and CT sets in the main utility branch that feed voltage and current information to the protective relay that houses the ANSI devices and triggers the circuit breaker in case of faults. They are next to each other and aren't past the transformer, as the transformer impedance and phase shift can change the readings. I also have PTs and CTs in the main branches of the BESS and Solar Power between the inverters and transformers, to give current and voltage information to the protective relay and in case of a fault in the 4.16kV busbar or an issue with the inverter, resulting in anti-islanding.

Equipment choices:

Used standard equipment - Equipment table in the schematic:
All equipment and impedance are based on IEEE and industry standards.

Transformers:
The transformers are mostly all 3-phase delta-wye transformers. The delta-wye transformers provide crucial safety features such as ground fault protection and fault isolation through the wye not being able to pass through the delta. They only require 3 wires from the delta primary side. The residential branch uses a single-phase split-phase transformer to drop 4.16kV to standard 120/240V household levels. Has 3 wires and allows for two different voltage outputs. One between the hot wire and neutral and another between two hot wires. This is due to the 180-degree phase shift

Inverters:
This design uses a 3-phase grid-tied inverter, as this is the industry standard for converting solar panel DC voltage into AC to flow into the grid. This design also uses a 4-quadrant bidirectional inverter for the BESS, as the battery storage system stores and releases DC energy. As the grid only uses AC voltage, we convert this voltage both ways, which is allowed through the 4 quadrants, allowing power to be output both ways.

Impedance and Power:
All the Impedance and Power output adhere to the IEEE standards. There is some potential impedance tolerance in each component based on IEEE manufacturing standards.

Industry Standards/Sources:
This design is based on IEEE standards

Some standards used are:
IEEE 1547-2018
ANSI/IEEE C37.2-2008
ANSI C84.1-2020

Some Standards omitted in the design:
Hardware Design testing
Control and automation testing
IEEE 2030.7 & 2030.8
IEEE 1547.1-2020

PYTHON Simulation:

Still in Progress. Doing a steady-state power flow analysis and fault simulation.